

OPERATION INSTRUCTIONS

Target Board: Digilent Basys 3 (Xilinx Artix-7)

RUN BITSTREAM FILES ON BASYS BOARD

1. Open Vivado Hardware Manager.
2. Connect the Basys 3 board via USB and click Open Target > Auto Connect.
3. Select Optimized_Version.bit or Extended_Version.bit and click Program Device.
4. The board begins executing immediately after programming completes.

RESET BUTTON

Button: BTNC (center push button)

Behavior: Active LOW — press to reset, release to run.

When reset is asserted (BTNC pressed):

PC is forced to address 0, all registers clear to 0.

When released (BTNC released):

Processor starts executing from address 0 automatically.

LED SIGNAL MAPPING

General Design (6 LEDs active)

LED	Signal	Meaning
LED0	result[0]	R7 bit 0 (LSB)
LED1	result[1]	R7 bit 1
LED2	result[2]	R7 bit 2
LED3	result[3]	R7 bit 3 (MSB)
LED14	zero	ALU zero flag (ON = result is 0)
LED15	overflow	ALU overflow flag

Extended Design (9 LEDs active)

LED	Signal	Meaning
LED0	result[0]	R7 bit 0 (LSB)
LED1	result[1]	R7 bit 1
LED2	result[2]	R7 bit 2
LED3	result[3]	R7 bit 3 (MSB)
LED14	zero (Z)	Result of last ALU/logic op was 0

LED15	overflow (V)	Arithmetic overflow occurred
LED13	greater (GT)	Last CMP result: $A > B$
LED12	equal (EQ)	Last CMP result: $A = B$
LED11	lower (LT)	Last CMP result: $A < B$

Note: LEDs are active HIGH. Only one of GT/EQ/LT is ON at a time.

SEVEN-SEGMENT DISPLAY MAPPING

The 4-digit display shows the value of register R7. The rightmost digit (AN0) displays the decimal value of R7. Remaining digits are blanked when the value is a single digit.

Segment lines (active LOW — segment lights when signal = 0):

Segment	Signal	Description
seg[0]	A	Top
seg[1]	B	Top-right
seg[2]	C	Bottom-right
seg[3]	D	Bottom
seg[4]	E	Bottom-left
seg[5]	F	Top-left
seg[6]	G	Middle

Digit anodes (active LOW — digit ON when signal = 0):

Anode	Digit Position
an[0]	Rightmost digit
an[1]	Second from right
an[2]	Second from left
an[3]	Leftmost digit

EXTENDED DESIGN — ADDITIONAL CONTROLS

Control	Description
SW15 = 0	Auto mode — processor runs continuously at ~1 Hz
SW15 = 1	Single-step / instruction injection mode
BTNC	Reset — press once to reset the processor
BTNU	Single-step — press once to execute one instruction (SW15 must be UP)

SW0-SW13	Manual instruction injection — set switches to 14-bit instruction encoding, SW13 = MSB, SW0 = LSB. Press BTNU to execute.
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EXPECTED BEHAVIOR

General Design

The processor computes $1 + 2 + 3 = 6$ and stores the result in R7.

Expected steady-state output:

LED3=0, LED2=1, LED1=1, LED0=0 (binary 0110 = decimal 6)

7-segment display shows "6" on the rightmost digit.

LED4 (zero) and LED5 (overflow) remain OFF.

The processor loops indefinitely, holding R7 = 6.

Extended Design

The processor runs a demo sequence: MOVI, ADD, SUB, NEG, loop.

R7 cycles through values $9 > 14 > 2$ then restarts.

Flag LEDs (Z, V, GT, EQ, LT) update according to each instruction.

In single-step mode, values update only on each BTNC press.